Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 October 2026
Team ID	Gyanvi Jain
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Used to filter, transform, aggregate, and analyse race data.	Tableau Calculated Fields, SQL (optional), HTML, CSS	Calculated fields support KPI calculations and Top N filtering, while HTML and CSS are used to build the web interface.
2.	Database / Data Warehouse	Stores and organizes structured Formula 1 race data.	Excel / CSV	The dataset contains five sheets with approximately 13,750 combined rows of race results, drivers, teams, circuits, and championship standings.
3.	Data Ingestion	Loads data from source files into the analytical environment.	Tableau Data Connections	Connects the Excel or CSV datasets to Tableau for analysis and visualization.
4.	Data Transformation and Business Logic	Cleans, joins, filters, and prepares data for analysis.	Tableau Desktop, Tableau Extracts, Calculated Fields	Handles missing values, validates data types, joins race results with circuit data, and creates measures such as DNF Count and cumulative points.

5.	Data Visualization / Analytics	Creates interactive charts, KPIs, dashboards, and reports.	Tableau Desktop / Tableau Public	Produces nine visualizations covering driver performance, team rankings, DNF reliability, tyre usage, geographic participation, and podium achievements.
6.	Front-End / User Interface	Provides an interface for users to explore dashboard insights.	Tableau Public, HTML, CSS, Flask	Tableau Public publishes the interactive dashboard, while Flask and HTML templates can embed it in a web application.